MARYLEBONE ROSTER · LINKS DESIGNER

Proposed CEA Link — December 2026

How this rota would work from December 2026, and how it compares with today's link

Prepared 28 September 2026 · every figure is calculated from the rota, not entered by hand

PROPOSAL

By the Book

Every shift time new, built from the December staffing levels, with longer earlies and shorter lates

CODE	**BB-24-D7**
FINGERPRINT	**0f14abce**
FAMILY	By the Book
CREATED	8 September 2026

✓ **24** lines, one per person · ✓ **35h 00m** a week on average — exactly the contract · ✓ longest run **6** days (limit 13) · ✓ **0** rests under 12h · ✓ **0** fatigue factors present (today 5)

Five questions a manager asks first, in order. Every figure is worked out from the rota; **page 2 says how to read each one**, with today's link and the other proposals marked on a scale.

CAN IT BE RUN?

Yes

meets every hard limit

0 rests under 12h · longest run 6 days (limit 13) · contract exact

DOES IT MEET THE DECEMBER RULES?

5 of 11

rules met

the staffing levels, the ticket office, the floor and the handovers — page 7 · today's link meets 3 of 11

HOW TIRING IS IT?

0

fatigue factors present, of 25

advisory — present means worth a look, not a breach · today 5 · fewest of the others 0

DOES IT FOLLOW THE TRAINS?

31.9

how closely the floor follows the weekday trains — 0 is a perfect match

today 68.8 · Saturday 9.7 · Sunday 30.9 · best of the others 20.4

IS IT FAMILIAR?

0 of 19

shift times already worked today

19 new, listed on page 5 · 12 of 20 working weeks keep one shift time (today 7 of 16)

6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20) · most of the others 25%

4 cover weeks

lines 1, 7, 13, 19

evenly spaced, every 6 lines · today's are at lines 1, 7, 12, 17

2h 52m a week

typical week-to-week move

how much the start time typically changes from one week to the next — today's link moves 4h 00m a week

What this is

By the Book is a proposal for the CEA link on the December 2026 timetable: a 24-line rotation for 24 people, where today's link has 20. In short: every shift time new, built from the December staffing levels, with longer earlies and shorter lates.

Against today's link — the table below; page 2 explains each figure.

AT A GLANCE — GREEN IS BETTER THAN TODAY, AMBER IS WORSE	TODAY'S LINK	BY THE BOOK
December 2026 rules met	3 of 11	**5 of 11**
People working each day — weekday · Saturday · Sunday	11–12 · 10 · 8	**14 · 14 · 10**
How closely the floor follows the trains — weekday · Sat · Sun (lower is closer)	68.8 · 68.7 · 72.5	**31.9 · 9.7 · 30.9**
Fatigue factors on the ORR (Office of Rail and Road) list (of 25)	5	**0**
Most days worked in a row (limit 13)	9	**6**
Shortest rest between two duties (at least 12 hours)	12h 30m	**13h 35m**
Full weekends off	4 in 20	**6 in 24**
Different shift times	18	**19 — none already worked today**

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a union rep yet. The December 2026 staffing levels were given verbally and are not yet in writing.

HOW TO READ THE NUMBERS

What each figure means, and what good looks like

Every figure on this sheet is worked out from the rota. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 22 proposals as gold and grey ticks.

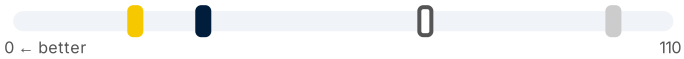
Three things to keep in mind. Lower is better on the fit, the fatigue count and the days in a row; higher is better on rest, weekends, weeks on one shift time and familiarity. “Present” is not “failed” — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets 3 of the 11 December 2026 rules.

DOES IT FOLLOW THE TRAINS?

How well the floor follows the trains — weekday

For each hour of an average weekday, the share of the day’s floor staff on duty is set against the share of the day’s trains — an hour with a tenth of the trains should have about a tenth of the staff. 0 would be a perfect match; a typical hour here is about 1.3 percentage points off, against 2.0 today. The ticket office is left out: it is staffed to its opening hours, not the trains.

What it does not tell you. It compares shares, not numbers: an extra person in a quiet hour makes the score worse though nobody is worse off. Read it beside the hour-by-hour numbers on page 6.



worst of the others **100** · best of the others (Right Away) **20.4**
today’s link **68.8** · this sheet **31.9**

Saturday **9.7** (today 68.7) · Sunday **30.9** (today 72.5). Everyone on duty, ticket office included: weekday 30.8 (today 51.1).

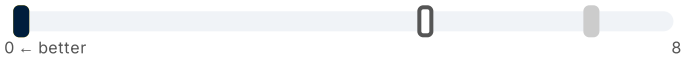
On page 6.

HOW TIRING IS IT?

Fatigue factors present

How many of the 25 roster patterns on the Office of Rail and Road’s good-practice list appear anywhere in the rotation — long runs of earlies, short gaps between duties, start times that jump — counted by the Links designer’s own module.

What it does not tell you. A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.



worst of the others **7**
best of the others (Eight Forty, Office Written In and 4 more) **0** · today’s link **5**
this sheet **0**

On page 8.

CAN IT BE RUN?

Most days worked in a row

The most days in a row anyone works in the worst case — a cover week’s four duties placed as badly as they can be. Chiltern’s limit is 13; six or fewer is good practice.

What it does not tell you. The roster clerk decides where a cover week’s four duties fall; page 7 says whether that changes this figure on this design.

worst of the others **9**
best of the others (Eight Forty, Evening Peak and 10 more) **6** · today’s link **9**
this sheet **6**

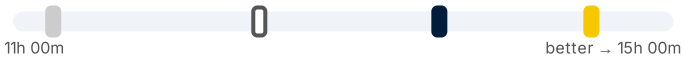
On page 7.

CAN IT BE RUN?

Shortest rest between two duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the minimum; a design with less cannot be run.

What it does not tell you. It is one gap. A rota with many rests of just over 12 hours and one with none look the same here; the number under 12 hours is at the top of page 1.



worst of the others **11h 15m**
best of the others (Anchored Lines, Frozen Block) **14h 30m**
today’s link **12h 30m** · this sheet **13h 35m**

On page 7.

WHAT IS IT LIKE TO WORK?

Full weekends off

The share of weeks with both Saturday and Sunday off. Today’s link gives 4 in 20, one weekend in 5; this sheet gives 6 in 24, one weekend in 4.

What it does not tell you. A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.

worst of the others **8%**
best of the others (Anchored Lines, Eight Forty and 10 more) **25%**
today’s link **20%** · this sheet **25%**

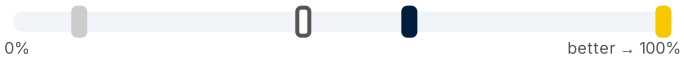
On page 5.

WHAT IS IT LIKE TO WORK?

Working weeks on one shift time

The share of working weeks where a person keeps the same clock time all week. A week that mixes an early and a late is what people mean by “all over the place”.

What it does not tell you. One shift time all week can still be a hard week if the shift is long; the week-to-week move and the duty lengths are on pages 5 and 8.



worst of the others **10%** · best of the others (Weekend Capped) **100%**
today’s link **44%** · this sheet **60%**

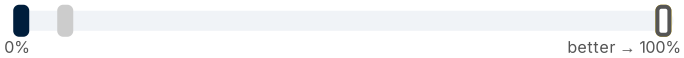
On page 5.

IS IT FAMILIAR?

Shift times people work today

The share of this sheet’s clock times that somebody on the 20-line link already works. It is a rough guide to how much there is to learn and get used to.

What it does not tell you. Familiar is not the same as popular, and a new time in one week is not the same as one in ten. The duty table on page 5 lists the new ones.



worst of the others **8%** · best of the others (Same Turns) **100%**
today’s link **100%** · this sheet **0%**

On page 5.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are fixed and which are yours to weigh, how much each figure can bear, and what no page here can tell you. Read this before the figures.

1 What this is
2 How to read the numbers
3 The decision it supports (this page)
4 The 24-line grid, with the words explained
5 A week, and the duty table
6 People on duty, hour by hour
7 Hard limits and the December 2026 rules
8 The ORR fatigue factors
9 Beside the other proposals
10 The rota, ready to paste into the app

The question

Should the CEA link for the December 2026 timetable be **By the Book**? It is judged against today's 20-line link and the December 2026 rules, and page 9 sets it beside the other 22 proposals on the same figures.

By the Book — every shift time new, built from the December staffing levels, with longer earlies and shorter lates.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

✓ Twelve hours between duties — met, tightest 13h 35m
✓ Chiltern's 13-day limit — longest run 6 days
✓ The 35-hour contract, exactly — 35h 00m

SOFT — WEIGH THESE AGAINST EACH OTHER

- December 2026 rules met: **5 of 11** (today 3 of 11)
- Fatigue factors present: **0** (today 5)
- How the floor follows the trains, weekday · Sat · Sun: **31.9 · 9.7 · 30.9** (today 68.8 · 68.7 · 72.5; lower is closer)
- Times worked today: **0 of 19** · full weekends off: **6** in 24 (today 4 in 20)

PREFERENCE — FOR STAFF TO SAY, NOT THE APP

- How long the shifts are: earlies 7h 00m–9h 30m, lates 7h 05m–8h 10m
- 19 of 19 clock times are new — the duty table on page 5 lists them
- The longest duty, 9h 30m — the grid on page 4 shows which lines carry it
- Cover weeks at lines 1, 7, 13, 19

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 14 on Saturday, 10 on Sunday	Worked out from the rota; checked again by the app when it is loaded	**Firm**
Longest run 6 days, inside 13	Computed; the written source of the 13-day limit is still to be confirmed	**Firm**
At least four at the open, three at the close, five at 22:00; 14 on Saturday, 10 on Sunday	December 2026 staffing levels given verbally; not yet in writing	**Provisional**
0 fatigue factors present	The ORR's 2021 list, which it states are not limits; 4 factors whose exact definition is still to be confirmed	**Advisory only** — never pass or fail
Fit, everyone on duty 30.8 · 8.1 · 26.3; floor only 31.9 · 9.7 · 30.9	The measured December 2026 timetable; one way of measuring, not the only one; the floor figure leaves the ticket office out	**A guide**
0 of 19 times are today's	Compared with today's 20-line link	**Firm**, but only a rough guide to what staff will accept

What no page here can tell you

- Whether staff accept the 19 shift times nobody works today (page 5 lists them) — this sheet counts times already worked, which is only a rough guide to what people will accept.
- Where the roster clerk places a cover week's four duties — on this design it makes no difference to any figure here.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 13, 19); nothing here models absence.
- Whether this pattern holds on Wembley event days — it is based on the train timetable, not on passenger numbers.

Before it is agreed

Every working line (all of them) hold at least one shift time nobody works today. On a rotating link everyone works every line in turn, so staff read those weeks first and say in one line each what they would change. Someone confirms the source of the 13-day limit. Until then, this sheet is an aid to discussion, not a decision.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON–SAT hours	ALL WEEK hours	DAYS WORKED Mon–Sat	TURNS Mon–Fri
1	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
2	15:15 23:25	16:00 23:55	RD	RD	15:45 23:55	15:45 23:55	16:00 23:55	32h 10m	40h 20m	4	2
3	14:15 22:25	16:15 23:55	RD	RD	16:15 23:55	16:15 23:55	RD	23h 00m	31h 10m	3	**1**
4	RD	14:15 22:00	14:15 22:00	14:15 22:00	14:15 22:00	14:15 22:00	RD	38h 45m	38h 45m	5	**1**
5	RD	07:15 16:30	07:15 16:30	07:15 16:30	06:20 15:45	06:20 15:45	RD	46h 35m	46h 35m	5	2
6	RD	15:45 23:55	15:45 23:55	15:45 23:55	RD	RD	16:15 23:55	32h 10m	32h 10m	4	**1**
7	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
8	07:15 16:45	07:15 16:45	RD	RD	06:20 15:30	07:15 16:45	06:20 15:30	37h 20m	46h 50m	4	2
9	07:15 15:30	06:20 13:30	06:20 13:30	RD	RD	06:20 13:30	07:15 16:45	31h 00m	39h 15m	4	**1**
10	08:30 18:00	RD	RD	06:20 13:30	06:20 13:30	06:20 15:30	RD	23h 30m	33h 00m	3	2
11	RD	15:15 22:45	15:15 22:45	15:15 22:45	15:15 22:45	RD	14:15 22:00	37h 45m	37h 45m	5	**1**
12	15:45 23:25	14:15 22:00	14:15 22:00	14:15 22:00	RD	RD	16:40 23:55	30h 30m	38h 10m	4	**1**
13	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
14	16:20 23:25	RD	RD	06:20 14:45	06:20 14:45	06:20 14:45	06:20 13:30	32h 25m	39h 30m	4	1
15	RD	RD	16:00 23:55	16:00 23:55	16:00 23:55	16:00 23:55	14:15 22:00	39h 25m	39h 25m	5	**1**
16	RD	RD	15:15 22:45	15:15 22:45	15:15 22:45	15:15 22:45	15:45 23:55	38h 10m	38h 10m	5	**1**
17	RD	RD	16:15 23:55	16:15 23:55	14:15 22:00	14:15 22:00	RD	30h 50m	30h 50m	4	2
18	RD	06:20 15:45	06:20 15:45	06:20 15:45	RD	RD	08:30 18:00	37h 45m	37h 45m	4	**1**
19	SP	SP	SP	SP	SP	SP	SP	*35h 00m*	*35h 00m*	—	—
20	14:15 22:25	15:15 22:45	RD	RD	07:15 16:30	07:15 16:30	06:20 14:45	34h 25m	42h 35m	4	2
21	07:15 16:30	06:20 14:45	06:20 14:45	RD	RD	15:15 22:45	15:15 22:45	31h 50m	41h 05m	4	2
22	RD	RD	07:15 16:45	07:15 16:45	07:15 16:45	07:15 16:45	RD	38h 00m	38h 00m	4	**1**
23	RD	07:15 16:45	07:15 16:45	07:15 16:45	07:15 16:45	RD	06:20 15:45	47h 25m	47h 25m	5	**1**
24	07:15 14:15	06:20 15:30	06:20 15:30	06:20 15:30	RD	RD	07:15 16:45	37h 00m	44h 00m	4	**1**
On duty	**10** SP 4	**14** SP 4	**14** SP 4	**14** SP 4	**14** SP 4	**14** SP 4	**14** SP 4	35h 00m	38h 27m	4.20	12 one turn
Minutes	4,965	7,000	7,000	7,000	7,000	7,000	7,000	**42,000**	**46,965**		

Early turns · Late turns · Cover (spare) week — four duties of seven, any turn · Rest day

Mon–Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: how many different shift times the week works Monday to Friday; **bold** marks a one-turn week (see below) · Minutes: total duty minutes each day — Monday to Saturday adds up to the contract (20 people × 35 hours = 42,000 minutes)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 11:00; a **late** starts at 11:00 or later.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

The floor — staff out on the station (gates, concourse, platforms), not in the ticket office.

Handover — the overlap when one shift takes over from another.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon–Sat and all week.

One-turn week — one clock time Monday to Friday, with every day of the week starting on the same side of 09:00 (shown bold in the Turns column); the easiest kind of week to live around.

Opener / closer — the first people on duty in the morning and the last at night.

Fit — how closely the number of people on duty follows the number of trains through the day; 0 is a perfect match, lower is better.

AGAINST TODAY'S LINK

A week, and the duty table

How a working week compares with today's, and every shift time beside the ones worked today.

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Weeks on one shift time	7 of 16	**12 of 20**
Weeks mixing early and late shifts	7	**3**
Rest-day breaks of two days or more (of all rest-day breaks)	13 of 17	**22 of 24**
Single rest days between duties (not a 48h break)	4	**2**
Days worked in a week (Sunday included; weeks × days)	1×3, 3×4, 11×5, 1×6	**6×4, 14×5**
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	**lines 1, 7, 13, 19**
Different shift times (19 not worked today)	18	**19**
Mon–Sat hours, average week (the contract)	35h 00m	**35h 00m**
Sunday duties (not contracted)	8 of 16 lines	**10 of 20 lines**

How many people are working

PEOPLE WORKING	TODAY	PROPOSED	WHY
Monday to Friday	11–12	**14**	24 people instead of 20, so more on each weekday
Saturday	10	**14**	meets the December 2026 figure of 14
Sunday	8	**10**	meets the December 2026 figure of 10
Ticket office Mon–Fri · Sat · Sun	4 · 4 · 3	**4 · 4 · 4**	ticket office only; today Sunday has one late, not two
Opening at 06:20 (07:15 Sunday) Mon–Fri · Sat · Sun	4 · 4 · 3	**4 · 4 · 4**	at least four to open, every day — met
Through to the close (23:55; 23:25 Sunday) Mon–Fri · Sat · Sun	2 · 2 · 3	**3 · 4 · 3**	at least three to close, every day — met
Still on duty at 22:00 Mon–Fri · Sat · Sun	4 · 2 · 3	**5 · 5 · 5**	at least five from 22:00, every day — met

The duty table, beside today's

	TODAY'S LINK			PROPOSED		
TURN	MON–FRI	SAT	SUN	MON–FRI	SAT	SUN
06:20-13:30 7h 10m · Early				1 (+1)	1 (+1)	
~~06:20-13:35 7h 15m · Early~~	~~1~~			~~0 (−1)~~		
~~06:20-13:45 7h 25m · Early~~	~~1~~			~~0 (−1)~~		
~~06:20-14:00 7h 40m · Early~~		~~1~~			~~0 (−1)~~	
~~06:20-14:20 8h 00m · Early~~	~~2~~			~~0 (−2)~~		
06:20-14:45 8h 25m · Early				1 (+1)	1 (+1)	
~~06:20-14:50 8h 30m · Early~~		~~3~~			~~0 (−3)~~	
06:20-15:30 9h 10m · Early				1 (+1)	1 (+1)	
06:20-15:45 9h 25m · Early				1 (+1)	1 (+1)	
07:15-14:15 7h 00m · Early						1 (+1)
07:15-15:30 8h 15m · Early						1 (+1)
~~07:15-15:45 8h 30m · Early~~			~~3~~			~~0 (−3)~~
07:15-16:30 9h 15m · Early				1 (+1)		1 (+1)
07:15-16:45 9h 30m · Early				2 (+2)	2 (+2)	1 (+1)
~~08:00-16:30 8h 30m · Early~~	~~2~~			~~0 (−2)~~		
~~08:30-16:30 8h 00m · Early~~			~~1~~			~~0 (−1)~~
08:30-18:00 9h 30m · Early					1 (+1)	1 (+1)
~~11:00-19:30 8h 30m · Late~~	~~1~~			~~0 (−1)~~		
~~12:00-20:00 8h 00m · Late~~		~~1~~			~~0 (−1)~~	
~~13:00-21:00 8h 00m · Late~~			~~1~~			~~0 (−1)~~
~~13:30-21:00 7h 30m · Late~~		~~1~~			~~0 (−1)~~	
~~13:30-22:00 8h 30m · Late~~	~~1~~			~~0 (−1)~~		
~~14:00-22:30 8h 30m · Late~~	~~2~~			~~0 (−2)~~		
14:15-22:00 7h 45m · Late				2 (+2)	2 (+2)	
14:15-22:25 8h 10m · Late						2 (+2)
~~14:30-22:00 7h 30m · Late~~		~~2~~			~~0 (−2)~~	
~~14:30-23:25 8h 55m · Late~~			~~3~~			~~0 (−3)~~
~~14:45-23:55 9h 10m · Late~~		~~2~~			~~0 (−2)~~	
15:15-22:45 7h 30m · Late				2 (+2)	1 (+1)	
15:15-23:25 8h 10m · Late						1 (+1)
~~15:15-23:55 8h 40m · Late~~	~~2~~			~~0 (−2)~~		
15:45-23:25 7h 40m · Late						1 (+1)
15:45-23:55 8h 10m · Late				1 (+1)	1 (+1)	
16:00-23:55 7h 55m · Late				1 (+1)	1 (+1)	
16:15-23:55 7h 40m · Late				1 (+1)	1 (+1)	
16:20-23:25 7h 05m · Late						1 (+1)
16:40-23:55 7h 15m · Late					1 (+1)	

How many people work each shift time on the busiest weekday, the Saturday and the Sunday. Green: more than today; amber: fewer; a struck-through time is one the proposal does not use.

Monday to Friday add up to 35,000 minutes of duty and Saturday 7,000 — together exactly 20 people × 35 hours = 42,000 minutes, the contract. Sunday adds 4,965, paid on top of the contracted week. 19 of the 19 shift times are new — nobody works them today.

COVER AGAINST THE SERVICE

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's trains.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday staffed 06:20–23:55 · a row for each different weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50.0
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2	2		
ON THE FLOOR MON · WED		2	2	4	4	4	5	5	5	3	5	5	3	3	3	2	2	2	2	83.7
ON THE FLOOR TUE · THU · FRI		2	2	4	4	4	5	5	6	4	6	6	4	4	4	3	3	2	2	64.4
PROPOSED		4	7	7	7	7	7	7	7	8	10	10	7	7	7	7	7	5	3	30.8
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2	2		
ON THE FLOOR		2	5	5	5	5	5	5	5	4	8	8	5	5	5	5	5	3	3	31.9
TRAINS, DEC 2026	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday staffed 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2			
ON THE FLOOR		2	2	2	2	2	2	3	4	5	4	4	4	4	4	3	2	2	2	68.7
PROPOSED		4	6	7	7	7	7	7	7	8	9	10	8	7	7	7	7	5	4	8.1
IN THE TICKET OFFICE		2	2	2	2	2	2	2	2	4	2	2	2	2	2	2	2			
ON THE FLOOR		2	4	5	5	5	5	5	5	4	7	8	6	5	5	5	5	5	4	9.7
TRAINS, DEC 2026	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday staffed 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
IN THE TICKET OFFICE			2	2	2	2	2	2	2	3	3	1	1	1	1	1	1	1		
ON THE FLOOR			1	2	2	2	2	2	3	5	5	4	3	3	3	3	2	2	3	72.5
PROPOSED			4	5	5	5	5	5	5	7	8	8	6	5	5	5	5	5	3	26.3
IN THE TICKET OFFICE			2	2	2	2	2	2	4	4	4	2	2	2	2	2	2	2		
ON THE FLOOR			2	3	3	3	3	3	1	3	4	6	4	3	3	3	3	3	3	30.9
TRAINS, DEC 2026			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Ticket office. Its staff are not floor cover. Monday to Saturday the office has two early and two late staff today, exactly as the December 2026 plan asks. **Sunday differs:** today there are two earlies and **one** late (a 14:30–23:25 closer, in the office until 22:30), where the plan has two and two. This design does not mark who works in the ticket office, so its floor figure is everyone on duty minus the four office staff the plan asks for. The indented rows do this for today and for the proposal, so the floor rows compare like with like.

Cover weeks are not in these figures — a cover week has no fixed times — so real cover is a little higher. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better. Page 1's weekday figure is worked out from the five weekdays combined, so it is not an average of the scores shown. On Sundays five trains in the December 2026 timetable arrive or leave after the 23:25 finish (the last at 23:54); the bar under the Sunday 23:00 hour marks the time after the finish. Whether Sunday cover should run later is still to be decided.

Reading it. The **floor** rows are the fair comparison. The ticket office is staffed to its opening hours, not to the trains, so counting it in can make a day look better or worse than the floor really is. The fit measures **shape, not numbers** — each hour's share of the day's staff against its share of the day's trains — so an extra person in a quiet hour makes it worse even though nobody is worse off. Read it beside the numbers of people, never alone.

THE RULES IT IS ASSESSED AGAINST

Hard limits and the December 2026 rules

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

✓ **No more than 13 days worked in a row** — longest possible run **6** days (today 9)

Longest possible run is 6 days, within Chiltern's limit of 13. A cover week is 4 duties in 7 days and the roster clerk places them, so this is the worst case the link allows. Worked as a block of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.
The written source of the 13-day limit is still to be confirmed.

✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today 0)

Tightest anywhere: **13h 35m — line 8 Fri 07:15-16:45 into Sat 06:20-15:30** (today's tightest: 12h 30m).

✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks

700h of duty a week across 20 working lines. Sundays (82h 45m) sit on top as RDW, as they do today. Individual weeks range 23h 00m–47h 25m; only the average is the contract.

The December 2026 rules — the same rules every proposal is measured against, with today's link beside them

RULE	TODAY'S LINK	THIS PROPOSAL	WHAT THE RULE ASKS
At least four on at the open, every day	✕ 4 · 4 · 3	✓ 4 · 4 · 4	from 06:20 (07:15 on a Sunday)
At least three through to the close, every day	✕ 2 · 2 · 3	✓ 3 · 4 · 3	until 23:55 (23:25 on a Sunday)
At least five still on duty at 22:00, every day	✕ 4 · 2 · 3	✓ 5 · 5 · 5	someone finishing at 22:00 does not count
Fourteen on a Saturday, ten on a Sunday	✕ 10 · 8	✓ 14 · 10	
Four cover weeks, evenly spread	✕ lines 1, 7, 12, 17	✓ lines 1, 7, 13, 19	evenly means every 6 lines, e.g. 1, 7, 13, 19
The ticket office: two early and two late, identical turns, every day	✕ Mon–Fri yes · Sat yes · Sun no	**✕ Mon–Fri no · Sat no · Sun no**	Mon–Fri 06:20–14:20 and 14:00–22:30; Sat 06:20–14:50 and 14:30–22:00; Sun two from 07:15 and two to 22:30
Every weekday closer starts at 15:45	✕ closers start at 15:15	**✕ closers start at 15:45, 16:00 and 16:15**	
At least two on the floor at every moment	✕ fewest 2 · 2 · 1	**✕ fewest 2 · 2 · 1**	ticket office staff not counted; checked every five minutes
Handovers: 15 minutes to each closer, 20 in the ticket office	✓ floor met · ticket office 20 min	**✕ floor not met · ticket office not in pairs**	on a Sunday each opener also stays until 15 minutes after the last closer arrives
Sunday duties between 8h and 9h	✓ 8h 00m–8h 55m	**✕ 7h 00m–9h 30m**	
No more shift times than today	✓ 18	**✕ 19**	today 18

Still to settle

- **6 of the 11 December 2026 rules are not met** — The ticket office: two early and two late, identical turns, every day; Every weekday closer starts at 15:45; At least two on the floor at every moment; Handovers: 15 minutes to each closer, 20 in the ticket office; Sunday duties between 8h and 9h; No more shift times than today. The shaded rows above give the figures.
- **New shift times.** 19 of the 19 are times nobody works today; page 5 lists them.
- **Sunday's finish.** Five trains in the December 2026 timetable arrive or leave after the 23:25 Sunday finish; whether Sunday cover should run later is still to be decided.

THE RULES IT IS ASSESSED AGAINST · CONTINUED

ORR fatigue factors

From the ORR's good-practice guidance, Fatigue Factors, page 3 (December 2021). ⚠ present — the pattern is in this design and worth a look, not a breach · ✓ clear — it is not · ● standing — comes with the station's hours or with any rotating link, not with this design's choices · a blank means it does not apply here.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlies, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	–	–
FF2	Early shift starting 05:00–07:00 Time of day	● 24	● 24
FF3	Very early shift starting before 05:00 Time of day	✓ 0	✓ 0
FF4	Very early shift (before 05:00) over 8h Duty length	–	–
FF5	Day shift over 12h Duty length	✓ 0	✓ 0
FF7	Early shift over 10h Duty length	✓ 0	✓ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	⚠ 3	✓ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	✓ 12 ✓ 9 as rostered	✓ 9 ✓ 9 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	✓ 0	✓ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	⚠ 6	✓ 4
MRSF	More than 12 consecutive day shifts Cumulative	✓ 9	✓ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	⚠ 60.8	✓ 54.4
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	⚠ 9	✓ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	–	–
FF17	Backward rotating pattern (definition to confirm) Circadian	✓ 10 backward / 12 forward	✓ 10 backward / 12 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	● 20-line rotation · typically 4h 00m a week	● 24-line rotation · typically 2h 52m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	⚠ 6	✓ 0
FF6	Night shift over 10h Night shifts	–	–
FF8	Less than 2 days rest after consecutive nights Night shifts	–	–
FF9	Less than 14h rest in any 24h for night shifts Night shifts	–	–
FF12	Only one day rest after night shifts Night shifts	–	–
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	–	–
FF16	More than 3 consecutive night shifts over 8h Night shifts	–	–
FF20	First night shift Night shifts	–	–
FF13	Less than 12h rest in any 24h Intervals between duties	✓ 0	✓ 0

Present: 0 (today 5). **How to read the table.** The figure beside each mark is the worst case found — a count of duties, a run of shifts, or hours in seven days. The MRSF rows are extra checks from the rail industry's guidance on managing staff fatigue, listed beside the ORR's own. FF2 applies to every 06:20 start, so it comes with the station's opening time, not with the design. FF11 counts shifts between two-day breaks, so a single rest day does not reset it — that is why it can be longer than the most days worked in a row. **Cover weeks.** On this design it makes no difference how the roster clerk places a cover week's four duties: the figures are the same either way. This page is an aid to discussion, not a fatigue risk assessment.

BESIDE THE OTHERS

Where it stands among the 23

Every proposal on the same figures, with today's link at the foot. This sheet's row is highlighted.

Like with like. Every figure below was worked out the same way for every proposal, from its rota, by the Marylebone Roster app — so any two rows can be compared directly.

PROPOSAL	RULES MET	FATIGUE FACTORS	MOST DAYS IN A ROW	SHORTEST REST	FULL WEEKENDS OFF	FLOOR FIT			SHIFT TIMES	ALREADY WORKED TODAY
						MON–FRI	SAT	SUN		
Right Away	11 of 11	0	6	14h 05m	6	20.4	11.6	19.2	17	5
Pinned Turns	8 of 11	0	6	14h 05m	6	30.9	31.7	58.1	16	9
Round Times	8 of 11	0	6	14h 05m	6	33.1	30.8	58.1	14	6
Quarter To	8 of 11	0	6	13h 05m	6	54.9	48.7	100.9	13	12
Weekend Capped	7 of 11	1	6	13h 35m	6	54.9	30.9	65.7	11	9
Same Turns	7 of 11	1	6	13h 30m	6	57.5	48.7	100.9	15	15
By the Book	**5 of 11**	**0**	**6**	**13h 35m**	**6**	**31.9**	**9.7**	**30.9**	**19**	**0**
Eight Forty	5 of 11	0	6	14h 20m	6	100.0	23.4	19.8	26	2
Office Written In	4 of 11	0	6	13h 50m	5	96.6	12.7	69.1	26	4
Gates Mended	4 of 11	1	6	14h 05m	6	33.3	45.4	39.3	15	4
Light Retime	4 of 11	3	9	12h 30m	4	37.2	31.1	47.0	23	11
Saturday Four	4 of 11	3	9	13h 35m	5	37.2	30.7	67.9	18	10
Weekday Lates	4 of 11	5	9	13h 35m	6	41.4	19.0	67.9	18	13
Fifteen Turns	4 of 11	7	9	11h 15m	2	33.3	47.3	39.3	15	5
Anchored Lines	3 of 11	1	7	14h 30m	6	38.1	19.0	67.9	19	13
Frozen Block	3 of 11	1	6	14h 30m	4	38.1	19.0	67.9	19	13
Evening Peak	3 of 11	1	6	13h 05m	6	41.9	19.0	67.9	19	13
Tenth Sunday	3 of 11	2	6	12h 30m	5	37.2	31.1	47.0	24	11
Cover at Seventeen	3 of 11	3	9	13h 35m	5	37.2	19.0	67.9	19	13
Short Closer	3 of 11	3	9	13h 35m	4	37.2	46.3	67.9	21	12
Targeted Fatigue Redo	3 of 11	4	9	13h 35m	5	37.2	19.0	67.9	19	13
Three Mondays	3 of 11	4	9	13h 35m	5	37.2	19.0	67.9	19	13
Weeks 17-18 Swapped	3 of 11	5	9	13h 35m	6	37.2	19.0	67.9	19	13
Today's 20-line link	3 of 11	5	9	12h 30m	4	68.8	68.7	72.5	18	18

Ordered by rules met, then fewest fatigue factors, then the weekday floor fit — an order, not a verdict. **Rules met** counts the 11 December 2026 rules on page 7. **Fatigue factors** are the ORR patterns present, of 25 (page 8). **Most days in a row** is at worst, against Chiltern's limit of 13; **shortest rest** must be at least 12 hours, or the rota cannot be run. **Full weekends off** are out of 24 (20 for today's link). **Floor fit** is how closely the people on the floor, ticket office staff not counted, follow the trains — lower is closer (page 6). **Shift times** is how many different times the rota uses, and **already worked today** how many of them people work now.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Paste the whole block into the app's Links designer (Links → Import). It re-runs every check on these pages, so nothing here has to be taken on trust.

The same rota is also supplied as two files, one to paste and one the app can open.

1	SP	SP	SP	SP	SP	SP	SP
2	15:15-23:25	16:00-23:55	RD	RD	15:45-23:55	15:45-23:55	16:00-23:55
3	14:15-22:25	16:15-23:55	RD	RD	16:15-23:55	16:15-23:55	RD
4	RD	14:15-22:00	14:15-22:00	14:15-22:00	14:15-22:00	14:15-22:00	RD
5	RD	07:15-16:30	07:15-16:30	07:15-16:30	06:20-15:45	06:20-15:45	RD
6	RD	15:45-23:55	15:45-23:55	15:45-23:55	RD	RD	16:15-23:55
7	SP	SP	SP	SP	SP	SP	SP
8	07:15-16:45	07:15-16:45	RD	RD	06:20-15:30	07:15-16:45	06:20-15:30
9	07:15-15:30	06:20-13:30	06:20-13:30	RD	RD	06:20-13:30	07:15-16:45
10	08:30-18:00	RD	RD	06:20-13:30	06:20-13:30	06:20-15:30	RD
11	RD	15:15-22:45	15:15-22:45	15:15-22:45	15:15-22:45	RD	14:15-22:00
12	15:45-23:25	14:15-22:00	14:15-22:00	14:15-22:00	RD	RD	16:40-23:55
13	SP	SP	SP	SP	SP	SP	SP
14	16:20-23:25	RD	RD	06:20-14:45	06:20-14:45	06:20-14:45	06:20-13:30
15	RD	RD	16:00-23:55	16:00-23:55	16:00-23:55	16:00-23:55	14:15-22:00
16	RD	RD	15:15-22:45	15:15-22:45	15:15-22:45	15:15-22:45	15:45-23:55
17	RD	RD	16:15-23:55	16:15-23:55	14:15-22:00	14:15-22:00	RD
18	RD	06:20-15:45	06:20-15:45	06:20-15:45	RD	RD	08:30-18:00
19	SP	SP	SP	SP	SP	SP	SP
20	14:15-22:25	15:15-22:45	RD	RD	07:15-16:30	07:15-16:30	06:20-14:45
21	07:15-16:30	06:20-14:45	06:20-14:45	RD	RD	15:15-22:45	15:15-22:45
22	RD	RD	07:15-16:45	07:15-16:45	07:15-16:45	07:15-16:45	RD
23	RD	07:15-16:45	07:15-16:45	07:15-16:45	07:15-16:45	RD	06:20-15:45
24	07:15-14:15	06:20-15:30	06:20-15:30	06:20-15:30	RD	RD	07:15-16:45

Prepared 28 September 2026. The code **0f14abce** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the Links designer shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was calculated by the Marylebone Roster app (version 24.35) from exactly these cells; paste them into the app and it shows the same figures.